05-30 第一篇

Subway Engineering: Building Underground Railways

Subways are one of the most important forms of urban transport. They move millions of people every day, reduce traffic above ground, and cut down on pollution. Yet building a subway is one of the hardest engineering tasks in the world. [1] That is because workers must dig through rock, soil, and sometimes water deep underground. They use huge machines called tunnel boring machines（盾构机）, or TBMs, to carve out tunnels. Each city has different conditions—some have soft soil, others have hard rock. Engineers must study the ground carefully before they start. One mistake can cause delays, extra cost, or even danger. In short, subway construction requires skill, patience, and advanced technology. It can take years or even decades to complete a single line.

How do engineers plan a subway line? [2] In other words, they study where people live, work, and travel. Planners look at maps of the city and count how many people use buses or cars on each route. They choose stations where many people can easily get on and off. They also avoid digging under important buildings or ancient ruins（遗址）when possible. Computers help engineers model traffic flow and design the best path. Public meetings let citizens share their opinions. The goal is to connect the most people with the fewest stations while keeping costs reasonable. Good planning can save time and money for years to come.

What happens during construction? [3] That is why building a subway affects daily life above ground. Roads may be closed or narrowed. Bus routes might change. Noise and dust can bother nearby residents. Construction can last many years. To reduce trouble, workers often build at night when fewer people are around. They use barriers to reduce noise and cover dig sites to keep dust down. Cities inform the public about schedule changes. For this reason, subway construction is a balance between progress and patience. When the line finally opens, many people forget the inconvenience. A new subway makes the city better for everyone.

How are subways kept safe? [4] It is true that safety systems run 24 hours a day. Trains have automatic brakes that stop them if something is on the track. Stations have cameras, emergency exits, and fire systems. Workers check tracks and tunnels regularly. In case of power failure, backup systems keep lights and fans running. During floods or earthquakes, subways can be closed until it is safe to run again. Safety is always the top priority. Engineers and operators train for emergencies so they can protect passengers. A safe subway is a reliable subway. Cities invest heavily in keeping their subways secure.

[5] In conclusion, subway engineering is a remarkable achievement. It brings together geology, machinery, and urban planning. From the first shovel to the first ride, building a subway is a long and complex journey. So the next time you take the subway, remember:

beneath your feet is years of work by thousands of engineers and workers. Let us appreciate the invisible effort that keeps our cities moving.

选项

A. That is because workers must dig through rock, soil, and sometimes water deep underground. B. In other words, they study where people live, work, and travel. C. That is why building a subway affects daily life above ground. D. It is true that safety systems run 24 hours a day. E. In conclusion, subway engineering is a remarkable achievement. F. For example, London opened the world's first subway in 1863.

答案键（Answer Key）

[1] A [2] B [3] C [4] D [5] E

深度解析

- **[1] A**：
  - **线索**：总领地铁建设的难度与挑战。空前描述"Subways are one of the most important forms of urban transport...Yet building a subway is one of the hardest engineering tasks in the world"，空后说明"Workers must dig through rock, soil, and sometimes water deep underground...They use huge machines called tunnel boring machines...Each city has different conditions...One mistake can cause delays, extra cost, or even danger...subway construction requires skill, patience, and advanced technology"。A选项"That is because workers must dig through rock, soil, and sometimes water deep underground"（那是因为工人必须在地下深处穿越岩石、土壤有时还有水体挖掘）与空后内容高度重合，"That is because"解释建设难度，衔接顺畅。
- **[2] B**：
  - **线索**：地铁线路的规划依据——居民出行与用地。空前设问"How do engineers plan a subway line?"，空后列举"Planners look at maps of the city and count how many people use buses or cars on each route...They choose stations where many people can easily get on and off...They also avoid digging under important buildings or ancient ruins...Computers help engineers model traffic flow...Public meetings let citizens share their opinions"。B选项"In other words, they study where people live, work, and travel"（换言之，他们研究人们居住、工作和出行的地点）中"In other words"概括空后方法，"where people live, work, and travel"与空后count people use buses、choose stations、traffic flow完美呼应，衔接自然。
- **[3] C**：

- **线索**：地铁施工对地面生活的影响。空前设问"What happens during construction?"，空后详述"Roads may be closed or narrowed. Bus routes might change. Noise and dust can bother nearby residents. Construction can last many years...To reduce trouble, workers often build at night...subway construction is a balance between progress and patience"。C选项"That is why building a subway affects daily life above ground"（因此修建地铁会影响地面上的日常生活）中"That is why"承接空前及空后结论，"affects daily life above ground"与空后roads closed、bus routes change、noise and dust精准对应，逻辑紧密。

- **[4] D**：
  - **线索**：地铁的安全保障——全天候安全系统。空前设问"How are subways kept safe?"，空后说明"Trains have automatic brakes...Stations have cameras, emergency exits, and fire systems...Workers check tracks and tunnels regularly...In case of power failure, backup systems keep lights and fans running...Safety is always the top priority"。D选项"It is true that safety systems run 24 hours a day"（的确，安全系统全天候运行）中"It is true that"引出安全机制，"safety systems run 24 hours a day"与空后automatic brakes、cameras、backup systems完美呼应，衔接得当。

- **[5] E**：
  - **线索**：全文收束与地铁工程的成就。空[5]位于结尾段段首，空后总结"subway engineering is a remarkable achievement...It brings together geology, machinery, and urban planning...beneath your feet is years of work by thousands of engineers and workers...the invisible effort that keeps our cities moving"。E选项"In conclusion, subway engineering is a remarkable achievement"（总之，地铁工程是一项非凡成就）用"In conclusion"总结全文，与空后首句完全一致，收束有力。

- **干扰项分析**：
  - **F. For example, London opened the world's first subway in 1863.**：涉及伦敦地铁历史，易被误填[1]或[2]。但F讨论的是历史案例，而空[1]侧重建设难度，空[2]侧重规划方法；F属"主题相关但论述层级错位"型干扰，与各空主旨不符。